

Interpretable Network Intrusion Detection

using Correlation-Based Feature Selection with a Hybrid CNN-LSTM Classifier

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Why this matters

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1

Computational cost

Deep IDS models require GPU training and high memory; full feature spaces of modern flow datasets are expensive to process.

2

Opacity

Analysts cannot easily explain why a deep model raised an alert. This blocks operational trust and conflicts with the GDPR right to explanation.

3

Class imbalance

In CICIDS2017, benign flows outnumber rare attacks (Heartbleed, Infiltration) by 3–5 orders of magnitude, biasing models towards majority class.

This project addresses all three concerns in a single end-to-end pipeline.

Research questions

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RQ1

Can correlation-based feature selection meaningfully reduce the 78-feature CICIDS2017 input space without harming detection accuracy?

RQ2

Does a hybrid CNN-LSTM model outperform a stacked LSTM baseline (the base paper), a Random Forest, and an XGBoost+SMOTE classifier on the same selected features?

RQ3

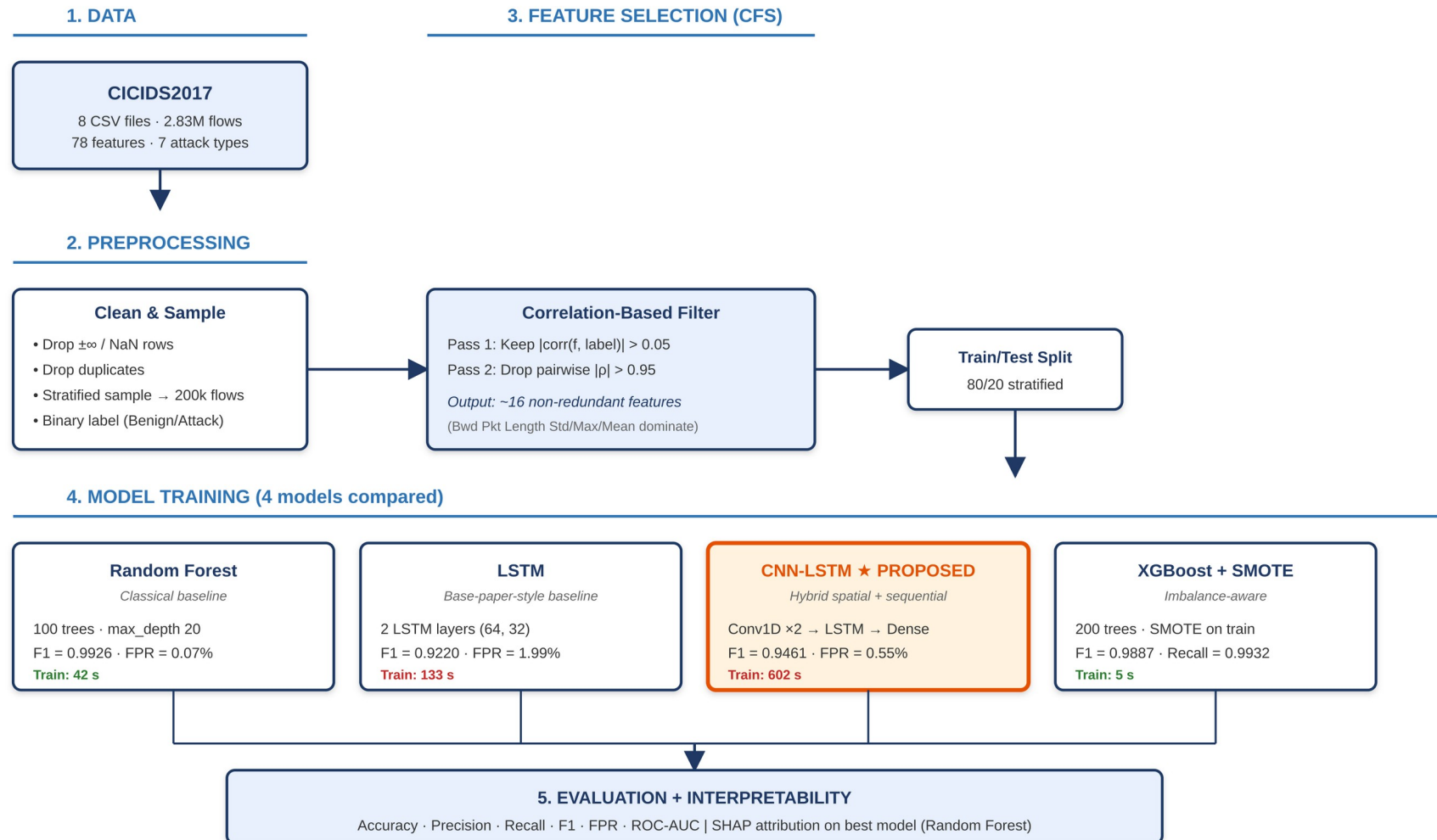
Which features drive the trained models' attack-class predictions, and can SHAP make those decisions interpretable enough to support analyst triage?

Three contributions: (1) reproducible CFS+4-model+SHAP pipeline, (2) honest comparison of hybrid vs classical baselines, (3) SHAP attribution on a modern benchmark.

System architecture

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System Architecture: Interpretable Network Intrusion Detection Pipeline



Dataset: CICIDS2017

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CICIDS2017 (Sharafaldin et al., 2018)

Canadian Institute for Cybersecurity

- 5 days of realistic enterprise traffic
- 2.83 million flows · 78 statistical features
- 7 attack categories: DoS, DDoS, brute force, port scan, web attack, infiltration, botnet
- **Severe class imbalance (3–5 orders of magnitude)**

Modelling decision

Binary task (Benign vs Attack) for tractable 4-day timeline; multi-class is direct extension.

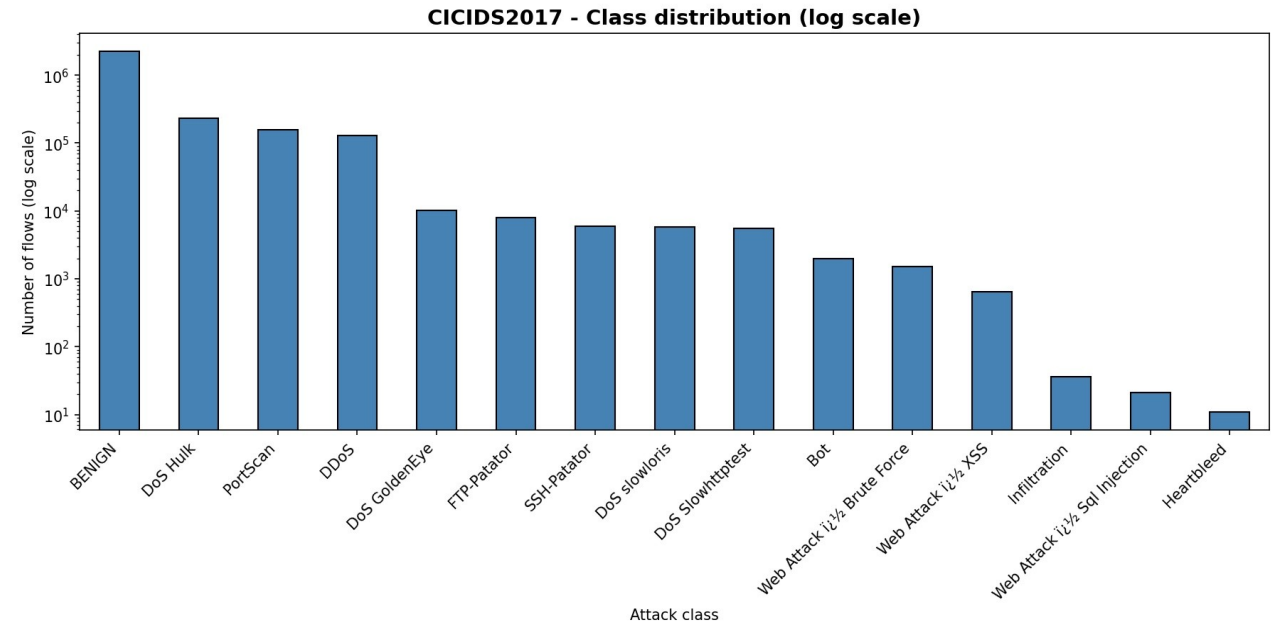


Figure 1 in report: log-scale class distribution. Benign flows dominate by 3–5 orders of magnitude.

200,000

stratified sample

80 / 20

train / test split

seed=42

reproducible

Correlation-based feature selection (CFS)

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Algorithm

Hall (2000): "A good feature subset is highly correlated with the class, but features within the subset are uncorrelated with each other."

Pass 1 — Class relevance

For each feature f , compute $|\text{corr}(f, \text{label})|$.
Keep features with $|\text{corr}| > 0.05$.

Pass 2 — Redundancy elimination

Sort kept features by $|\text{corr}|$ descending.
Walk the list; drop any feature whose pairwise $|\rho|$ with an already-kept feature exceeds 0.95.

Result: 78 → ~16 features (~80% reduction)

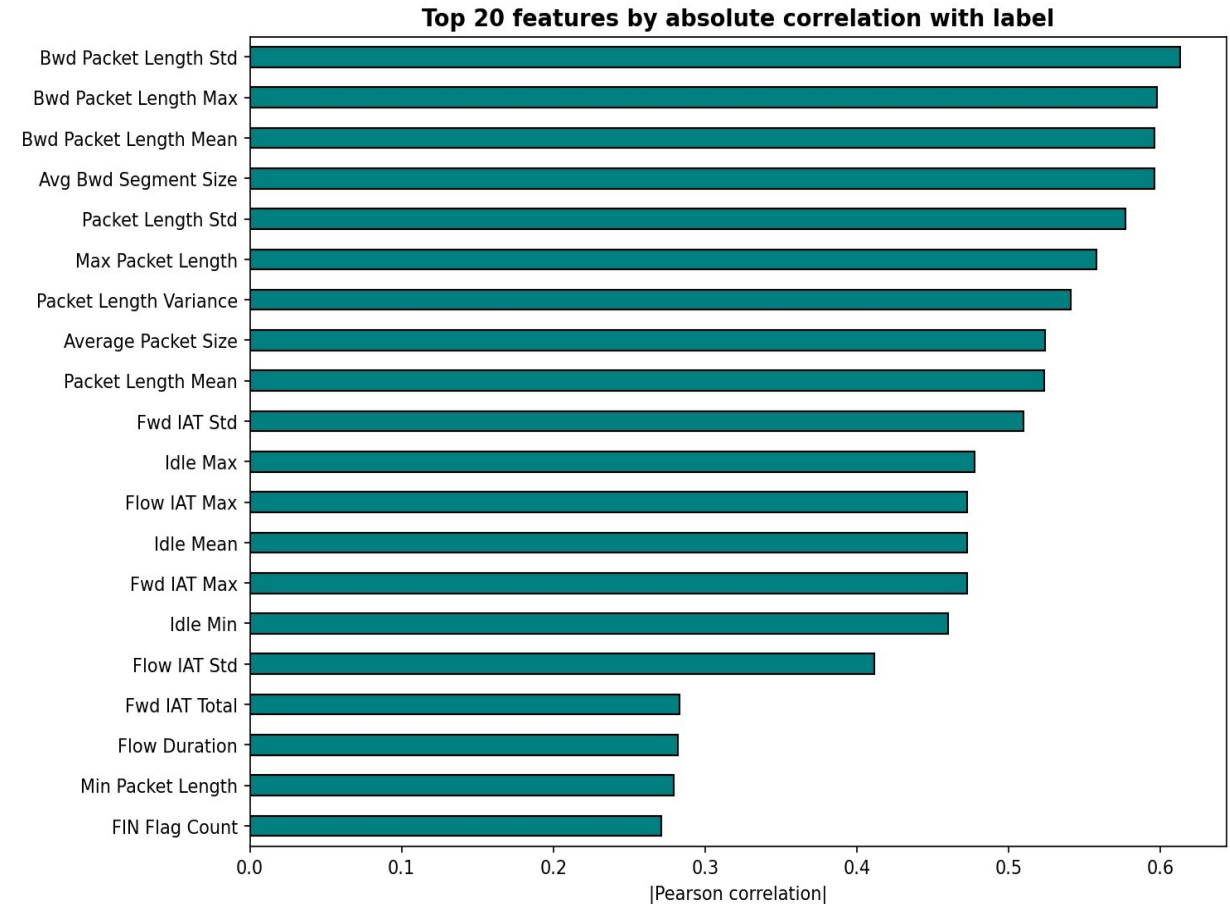


Figure 2 in report: top 20 features by |Pearson correlation| with attack label. Backward packet-length statistics dominate.

Four models compared

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Random Forest

Classical baseline

100 trees · max_depth 20 · scikit-learn

Train time

42 s

LSTM

Base-paper-style

2 LSTM layers (64, 32) · dropout 0.3

Train time

133 s



CNN-LSTM

PROPOSED (hybrid)

Conv1D ×2 → LSTM → Dense
Alsaiani & Ilyas (2025)

Train time

602 s

XGBoost + SMOTE

Imbalance-aware

200 trees · SMOTE on training fold
Li et al. (2023)

Train time

5 s

Rationale: span the methodological spectrum — classical ensemble · single-architecture deep · hybrid deep · imbalance-aware boosting.

Results — comparative metrics

Model	Acc	Prec	Recall	F1	FPR	AUC	Time
Random Forest	0.9975	0.9964	0.9889	0.9926	0.0007	0.9999	42 s
LSTM (base-paper style)	0.9732	0.9054	0.9392	0.9220	0.0199	0.9952	133 s
CNN-LSTM (proposed)	0.9823	0.9716	0.9218	0.9461	0.0055	0.9984	602 s
XGBoost + SMOTE	0.9962	0.9842	0.9932	0.9887	0.0032	0.9998	5 s

Best F1 + FPR

Random Forest

F1=0.9926 · FPR=0.07%

Best Recall + Speed

XGBoost+SMOTE

Recall=0.9932 · 5 s train

CNN-LSTM beats LSTM

+2.4 F1 pp · -3.6× FPR

Hybrid edge confirmed

Honest finding: classical and imbalance-aware methods edge out my proposed hybrid on this binary task.

Visual evaluation — ROC and confusion matrices

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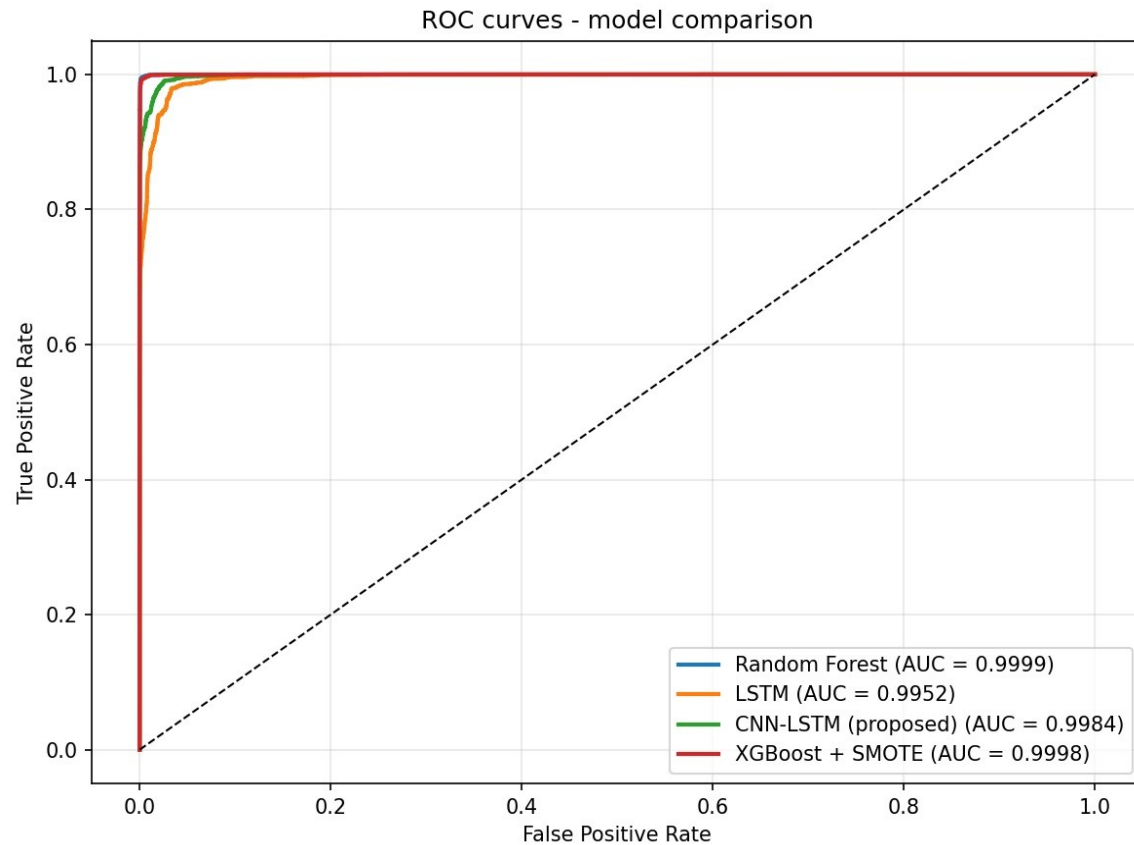


Figure 4: ROC curves. RF and XGBoost+SMOTE dominate the upper-left corner; CNN-LSTM clearly above plain LSTM.

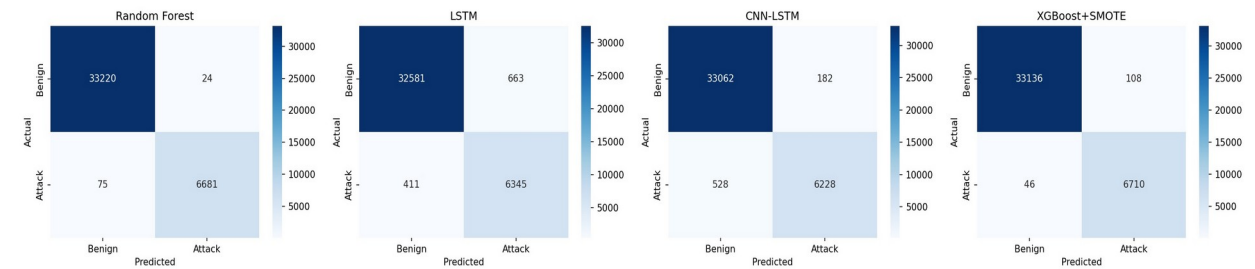


Figure 3: Confusion matrices for all four models on the test set.

Confusion matrix takeaways

- RF: only 24 false positives out of 33,244 benign flows
- LSTM: 663 FP and 411 FN — worst on both axes
- CNN-LSTM cuts LSTM's FPs to 182 (3.6× reduction)
- XGBoost+SMOTE: only 46 missed attacks (best recall)

Live demo: SHAP interpretability in Colab

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LIVE DEMO

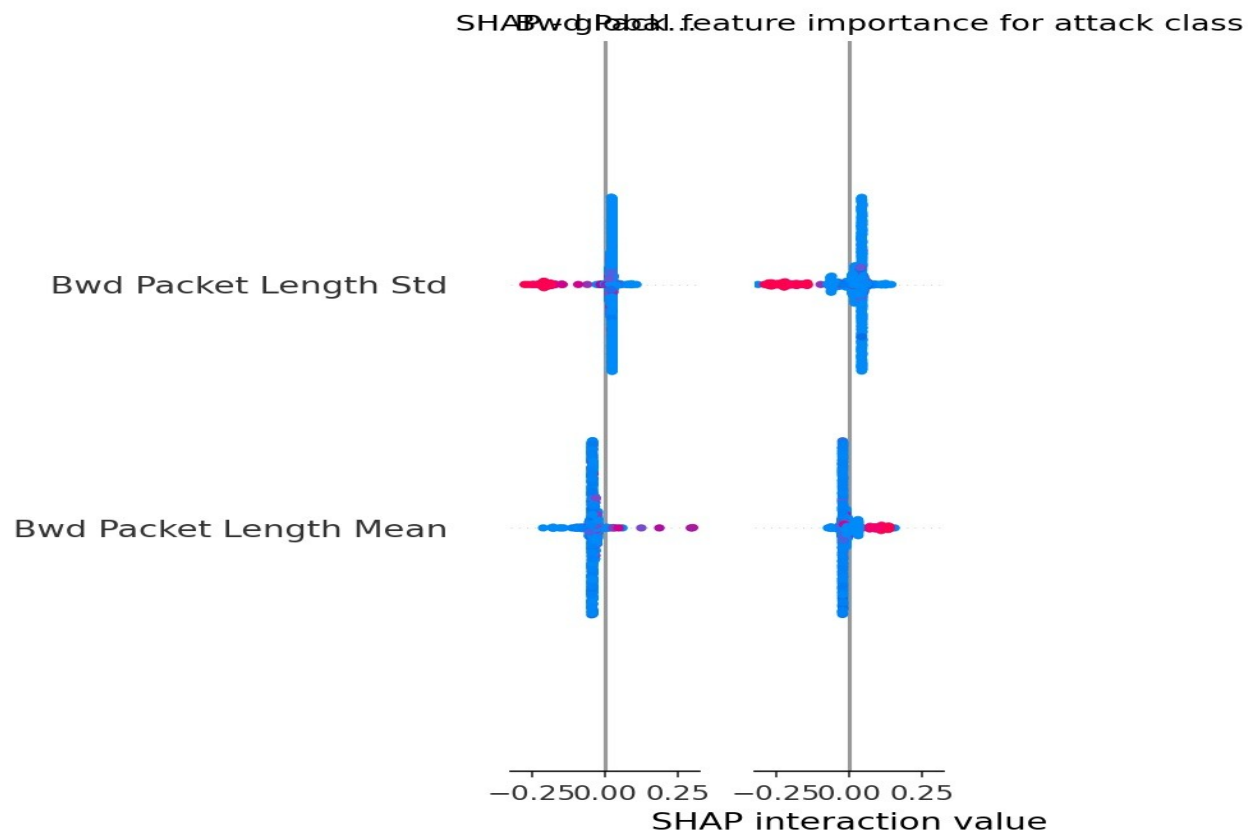
Switching to Google Colab now

What I'll show:

1. The CFS function and selected features (~10 s)
2. The CNN-LSTM model summary (~10 s)
- 3. Run the SHAP cell live (~30 s)**
4. Walk through the SHAP global importance plot (~30 s)

All cells are pre-run; I am only re-executing the SHAP cell.

Pre-computed reference (in case of demo issue):



Figures 5–6: SHAP attributions (Random Forest). Top features align with Pearson ranking from Slide 6.

Conclusions

- CFS reduced 78 → ~16 features without harming accuracy across all four classifiers.
- Proposed CNN-LSTM beats LSTM baseline on every metric (+2.4 F1 pp, -3.6× FPR).
- **But Random Forest and XGBoost+SMOTE are stronger overall — honest finding.**
- SHAP top features align with Pearson — audit trail for analysts.

Limitations

- Binary task only (multi-class deferred).
- 200k stratified sample, not full 2.83M.
- SHAP applied only to RF (TreeExplainer); KernelExplainer on deep models too costly.
- No cross-testbed validation (Apuzzese et al. 2022 caveat).
- No adversarial robustness testing.

Future work

- Multi-class extension on all 7 attack types.
- Add SMOTE / focal loss to deep models — likely to close the gap to XGBoost+SMOTE.
- KernelExplainer SHAP on CNN-LSTM for full interpretability story.
- Cross-testbed validation (CICIDS2017 → UNSW-NB15).
- Adversarial evasion testing.

Thank you. Questions?